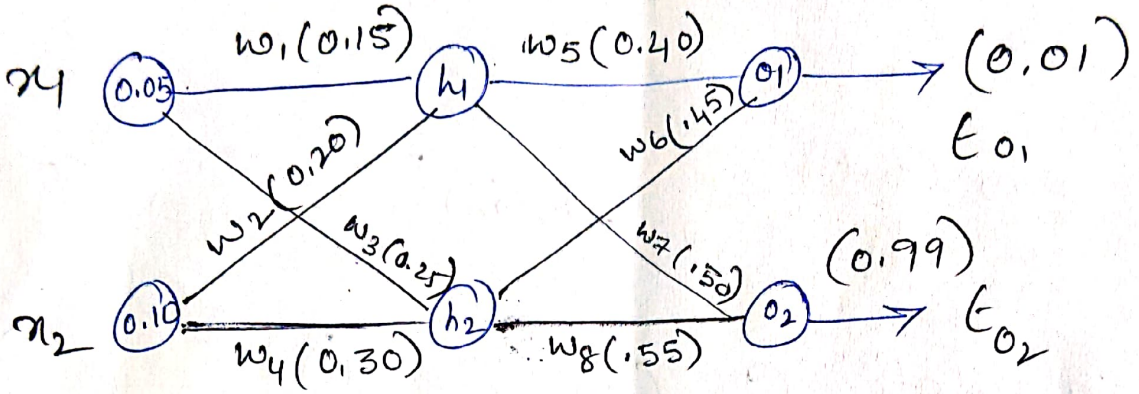
When error occurs, we go in backward direction
ie output → hidden → input Layer.

Part 1: Calculate Forward propagation Error



x_1, x_2 ---- inputs
w_1, w_2, w_3 ---- weights.
O_1, O_2 ---- outputs

we will calculate the o/p which we should actually get

Here bias factors $b_1 = 0.35$ b/w $x_i^o \rightarrow h_i^o$

$b_2 = 0.60$ b/w $h_i^o \rightarrow O_i^o$

Part-1 calculation

(1) calculate h_i (in and out)

$$h_1(in) = w_1 x_1 + w_2 x_2 + b_1$$

$$= .15 * 0.05 + 0.20 * 0.10 + 0.35$$

$$= 0.3777$$

$$h_1(out) = \frac{1}{1+e^{-h_i(in)}} = \frac{1}{1+e^{-0.3777}}$$

$$= 0.5932$$

(b) Calculate h_2 (in and out)

$$h_2(in) = w_3 x_1 + w_4 x_2 + b_2$$

$$= 0.25(0.05) + 0.3(0.1) + 0.35$$

$$= 0.125 + 0.03 + 0.35$$

$$= 0.3925$$

$$h_2(out) = \frac{1}{1+e^{-h_2(in)}} = \frac{1}{1+e^{(-.3925)}} = 0.5968$$

(c) Calculate O_1 (in & out)

$$O_1(in) = h_1(out) * w_5 + h_2(out) * w_6 + b_2$$

$$= 0.5932 * (0.40) + 0.5968(0.45) + 0.60$$

$$= 1.105$$

$$O(out) = \frac{1}{1+e^{-O_1(in)}} = \frac{1}{1+e^{-1.105}} = 0.7513$$

(d) Calculate O_2 (in & out)

$$O_2(in) = h_1(out) * w_7 + h_2(out) * w_8 + b_2$$

$$= 0.5932(0.50) + 0.5968(0.55) + 0.6$$

$$= 1.92484$$

$$O_2(out) = \frac{1}{1+e^{-O_2(in)}} = \frac{1}{1+e^{-1.92484}}$$

$$= 0.7729$$

we begin backward phase.

we adjust weights leading to O_1 & O_2

$\Delta w_{ji} = \eta \delta_j O_i$ $\quad \delta_j^o$ – Error at j^{th} unit

$O_i^o \rightarrow$ actual o/p calculated.

If Error at output Layer

$$\boxed{\delta_j^o = O_j^o (1 - O_j^o)(t_j - O_j^o)}$$

If Error at hidden Layer

$$\boxed{\delta_j = O_j^o (1 - O_j^o) \sum_k \delta_k w_{kj}}$$

$$\delta_{O_1} = (0.75137)(1 - 0.75137)(0.01 - 0.7513)$$

$$= -0.13850$$

$$w_5^* = w_5 + \eta \delta_{O_1} \times \text{out}_{h_1}$$

$$= w_5 + 0.5 \times (-0.13850) \times 0.59327$$

$$= 0.35892$$

$$w_6^* = w_6 + \eta \times \delta_{O_1} \times \text{out}_{h_2} \rightarrow 0.40867$$

$$= 0.45 + 0.5 \times (-0.13850) \times (0.59689)$$

$$b_3 = b_3 + \eta * \delta_{01} * 1$$
$$= 0.60 + 0.5 * (-0.13850) \times 1$$
$$= 0.53075$$

$$\delta O_2 = (0.77293)(1 - 0.77293)$$
$$(0.99 - 0.77795$$
$$= 0.03810$$

$$w_7^* = w_7 + \eta \delta_{O_2} * \text{out } h_1$$
$$= 0.50 + (0.5 * 0.03810) \times 0.5933$$
$$= 0.51130$$

$$w_8 = w_8 + \eta \delta_{O_2} * \text{out } h_2$$
$$= 0.55 + 0.5 \times 0.03810 *$$
$$0.59689.$$
$$= 0.56137$$

$$b_4^* = b_4 + \eta * \delta O_2 * 1$$
$$= 0.60 + 0.5 * 0.03810 * 1$$
$$= 0.61905$$

we begin backward phase we adjust the weights leading to O_1 & O_2

$\Delta w_{ji} = \eta \delta_j o_i$ $\delta_j \rightarrow$ Error term at j^{th} unit

$o_i \rightarrow$ actual o/p calculated.

$\delta_j = o_j (1-o_j)(t_j - o_j)$ $\delta_j = o_j (1-o_j) \sum_k \delta_k w_{kj}$

$\delta_{O1} = (0.75137) \times (1-0.75137) \times (0.01 - 0.75137)$

$= -0.13850$

$w_5^{\alpha} = w_5 + \eta \delta_{O1} \times out_{h1}$ (0.40)

$= w_5 + 0.5 \times (-0.13850) \times 0.59327$

$= 0.35892$

$w_6^{\alpha} = w_6 + \eta \alpha \delta_{O1} \alpha out_{h2}$

$= 0.45 + 0.5 \times (-0.13850) \alpha (0.59689)$

$= 0.40867$

$b_3 = b_3 + \eta \alpha \delta_{O1} \alpha 1$ (1 →)

$= 0.60 + 0.5 \alpha (-0.13850) \times 1$

$= 0.53075$

$\delta_{O2} = (0.77293)(1-0.77293)(0.99 - 0.77293)$

$= 0.03810$

$w_7^{\alpha} = w_7 + \eta \delta_2 \alpha out_{h1}$

$= 0.50 + 0.5 \alpha 0.03810 \times 0.59327$

$= 0.51130$

$\delta_{O_2} = 0.77293 \times (1 - 0.77293) \times (0.99 - 0.77293)$

$= 0.03810$

$w_7^* = w_7 + \eta * \delta_{O_2} \times \text{out}_{h_1}$

$= 0.50 + 0.5 \times 0.03810 \times 0.59327$

$= 0.51130$

$w_8^* = w_8 + \eta * \delta_{O_2} * \text{out}_{h_2}$

$= 0.55 + 0.5 \times 0.03810 \times 0.59689$

$= 0.56137$

$b_4^* = b_4 + \eta * \delta_{O_2} \times 1$

$= 0.60 + 0.5 \times 0.03810 \times 1$

$= 0.61905$

$\delta_{h_1} = \text{out}(h_1) \times (1 - \text{out}_{h_1})(\delta_{O_1} \times w_5 + \delta_{O_2} \times w_7)$

$=$

Thu, fri, sat
Thu, wed, thu
Fri, Saturday.

$$w_1^* = w_1 + \eta \delta_{h_1} \times i_1$$

$$= 0.15 + 0.5 \times (-0.00872) \times 0.05$$

$$0.14978$$

$$w_2^* = w_2 + \eta \delta_{h_1} \times i_2$$

$$= 0.20 + 0.5 \times (-0.00872) \times 0.10$$

$$= 0.19956$$

$$b_1^* = b_1 + \eta \times \delta_{h_1} \times 1$$

$$= 0.35 + 0.5 \times (-0.00572) \times$$

$$= 0.34562$$